

ENERGY MANAGEMENT AND AUDIT

Definition & Objectives of Energy Management

The fundamental goal of energy management is to produce goods and provide services with the **least cost and least environmental effect.**

"The judicious and effective use of energy to maximize profits (minimize costs) and enhance competitive positions"

(Cape Hart, Turner and Kennedy, Guide to Energy Management Fairmont press inc. 1997)

The objective of Energy Management is to achieve and maintain optimum energy procurement and utilization, throughout the organization and:

- To minimize energy costs / waste without affecting production & quality
- To minimise environmental effects.

Energy Audit: Types And Methodology

Energy Audit is defined as “the verification, monitoring and analysis of use of energy including submission of technical report containing recommendations for improving energy efficiency with cost benefit analysis and an action plan to reduce energy consumption”.

Energy Audit is the key to a systematic approach for decision-making in the area of energy management.

- It attempts to balance the total energy inputs with its use, and serves to identify all the energy streams in a facility.
- It quantifies energy usage according to its discrete functions.
- Industrial energy audit is an effective tool in defining and pursuing comprehensive energy management programme.

Need for Energy Audit

In any industry, the three top operating expenses are often found to be energy (both electrical and thermal), labour and materials. If one were to relate to the manageability of the cost or potential cost savings in each of the above components, energy would invariably emerge as a top ranker, and thus energy management function constitutes a strategic area for cost reduction. **Energy Audit will help to understand more about the ways energy and fuel are used in any industry, and help in identifying the areas where waste can occur and where scope for improvement exists.**

In general, Energy Audit is the translation of conservation ideas into realities, by lending technically feasible solutions with economic and other organizational considerations within a specified time frame.

The primary objective of Energy Audit is **to determine ways to reduce energy consumption per unit of product output or to lower operating costs.** Energy Audit provides a “bench-mark” (Reference point) for managing energy in the organization and also provides the basis for planning a more effective use of energy throughout the organization.

Type of Energy Audit

The type of Energy Audit to be performed depends on:

- Function and type of industry
- Depth to which final audit is needed, and
- Potential and magnitude of cost reduction desired

Thus Energy Audit can be classified into the following two types.

- i) Preliminary Audit
- ii) Detailed Audit

Preliminary Energy Audit Methodology

Preliminary energy audit is a relatively quick exercise to:

- Establish energy consumption in the organization
- Estimate the scope for saving
- Identify the most likely (and the easiest areas for attention
- Identify immediate (especially no-/low-cost) improvements/ savings
- Set a 'reference point'
- Identify areas for more detailed study/measurement
- Preliminary energy audit uses existing, or easily obtained data

Detailed Energy Audit Methodology

A comprehensive audit provides a detailed energy project implementation plan for a facility, since it evaluates all major energy using systems.

This type of audit offers the most accurate estimate of energy savings and cost. It considers the interactive effects of all projects, accounts for the energy use of all major equipment, and includes detailed energy cost saving calculations and project cost.

In a comprehensive audit, one of the key elements is the energy balance. This is based on an inventory of energy using systems, assumptions of current operating conditions and calculations of energy use. This estimated use is then compared to utility bill charges.

Detailed energy auditing is carried out in three phases: Phase I, II and III.

1. Phase I - Pre Audit Phase
2. Phase II - Audit Phase
3. Phase III - Post Audit Phase

Ten Steps Methodology for Detailed Energy Audit

Step No	PLAN OF ACTION	PURPOSE / RESULTS
Step 1	<u>Phase I –Pre Audit Phase</u> • Plan and organise • Walk through Audit • Informal Interview with Energy Manager, Production / Plant Manager	• Resource planning, Establish/organize a Energy audit team • Organize Instruments & time frame • Macro Data collection (suitable to type of industry.) • Familiarization of process/plant activities • First hand observation & Assessment of current level operation and practices
Step 2	• Conduct of brief meeting / awareness programme with all divisional heads and persons concerned (2-3 hrs.)	• Building up cooperation • Issue questionnaire for each department • Orientation, awareness creation

Step 3	<u>Phase II –Audit Phase</u> Primary data gathering, Process Flow Diagram, & Energy Utility Diagram	- Historic data analysis, Baseline data collection - Prepare process flow charts - All service utilities system diagram (Example: Single line power distribution diagram, water, compressed air & steam distribution. - Design, operating data and schedule of operation - Annual Energy Bill and energy consumption pattern (Refer manual, log sheet, name plate, interview)
Step 4	Conduct survey and monitoring	Measurements : Motor survey, Insulation, and Lighting survey with portable instruments for collection of more and accurate data. Confirm and compare operating data with design data.
Step 5	Conduct of detailed trials /experiments for selected energy guzzlers	- Trials/Experiments: 24 hours power monitoring (MD, PF, kWh etc.). - Load variations trends in pumps, fan

		compressors etc. - Boiler/Efficiency trials for (4 – 8 hours) - Furnace Efficiency trials Equipments Performance experiments etc
Step6	• Analysis of energy use	• Energy and Material balance & energy loss/waste analysis
Step 7	• Identification and development of Energy Conservation (ENCON) opportunities	• Identification & Consolidation ENCON measures ▪ Conceive, develop, and refine ideas ▪ Review the previous ideas suggested by unit personal ▪ Review the previous ideas suggested by energy audit if any ▪ Use brainstorming and value analysis techniques ▪ Contact vendors for new/efficient technology

Step 8	• Cost benefit analysis	• Assess technical feasibility, economic viability and prioritization of ENCON options for implementation • Select the most promising projects • Prioritise by low, medium, long term measures
Step9	• Reporting & Presentation to the Top Management	• Documentation, Report Presentation to the top Management.
Step10	<u>Phase III –Post Audit phase</u> • Implementation and Follow-up	Assist and Implement ENCON recommendation measures and Monitor the performance ▪ Action plan, Schedule for implementation ▪ Follow-up and periodic review

Phase I –Pre Audit Phase Activities

A structured methodology to carry out an energy audit is necessary for efficient working. An initial study of the site should always be carried out, as the planning of the procedures necessary for an audit is most important.

Initial Site Visit and Preparation Required for Detailed Auditing

An initial site visit may take one day and gives the Energy Auditor/Engineer an opportunity to meet the personnel concerned, to familiarize him with the site and to assess the procedures necessary to carry out the energy audit.

During the initial site visit the Energy Auditor/Engineer should carry out the following actions: -

- **Discuss** with the site's senior management **the aims of the energy audit**.
- **Discuss economic guidelines** associated with the recommendations of the audit.
- **Analyse the major energy consumption data** with the relevant personnel.
- **Obtain site drawings** where available – building layout, steam distribution, compressed air distribution, electricity distribution etc.
- **Tour the site** accompanied by engineering/production

The main aims of this visit are: -

- To **finalise Energy Audit team**
- To identify **the main energy consuming areas**/plant items to be surveyed during the audit.
- To identify **any existing instrumentation/ additional metering required**.
- To decide whether any meters will have to be installed prior to the audit eg. kWh, steam, oil or gas meters.
- To identify the instrumentation required for carrying out the audit.
- To plan with time frame
- To collect macro **data on plant energy resources**, major energy consuming centers
- **To create awareness through meetings/ programme**

Phase II- Detailed Energy Audit Activities

Depending on the **nature and complexity of the site**, a comprehensive audit can take from **several weeks to several months to complete.**

Detailed studies to establish, and investigate, energy and material balances for specific plant departments or items of process equipment are carried out. Whenever possible, **checks of plant operations are carried out over extended periods of time, at nights and at weekends as well as during normal daytime** working hours, **to ensure that nothing is overlooked.**

The **audit report** will include **a description of energy inputs and product outputs** by major department or by major processing function, and **will evaluate the efficiency of each step of the manufacturing process.**

Means of improving these efficiencies will be listed, and at least a preliminary assessment of the cost of the improvements will be made to indicate the expected payback on any capital investment needed.

The audit report should **conclude with specific recommendations for detailed engineering studies and feasibility analyses**, which must then be performed to justify the implementation of those conservation measures that require investments.

The information to be collected during the detailed audit includes: -

1. Energy consumption by type of energy, by department, by major items of process equipment, by end-use
2. Material balance data (raw materials, intermediate and final products, recycled materials, use of scrap or waste products, production of by-products for re-use in other industries, etc.)
3. Energy cost and tariff data
4. Process and material flow diagrams
5. Generation and distribution of site services (eg.compressed air, steam).
6. Sources of energy supply (e.g. electricity from the grid or self-generation)
7. Potential for fuel substitution, process modifications, and the use of co-generation
8. systems (combined heat and power generation).
9. Energy Management procedures and energy awareness training programs within the establishment.

Existing baseline information and reports are **useful to get consumption pattern, production cost and productivity levels in terms of product per raw material inputs**. The audit team should collect the following baseline data:

- Technology, processes used and equipment details
- Capacity utilisation
- Amount & type of input materials used
- Water consumption
- Fuel Consumption
- Electrical energy consumption
- Steam consumption
- Other inputs such as compressed air, cooling water etc
- Quantity & type of wastes generated
- Percentage rejection / reprocessing
- Efficiencies / yield

DATA COLLECTION HINTS

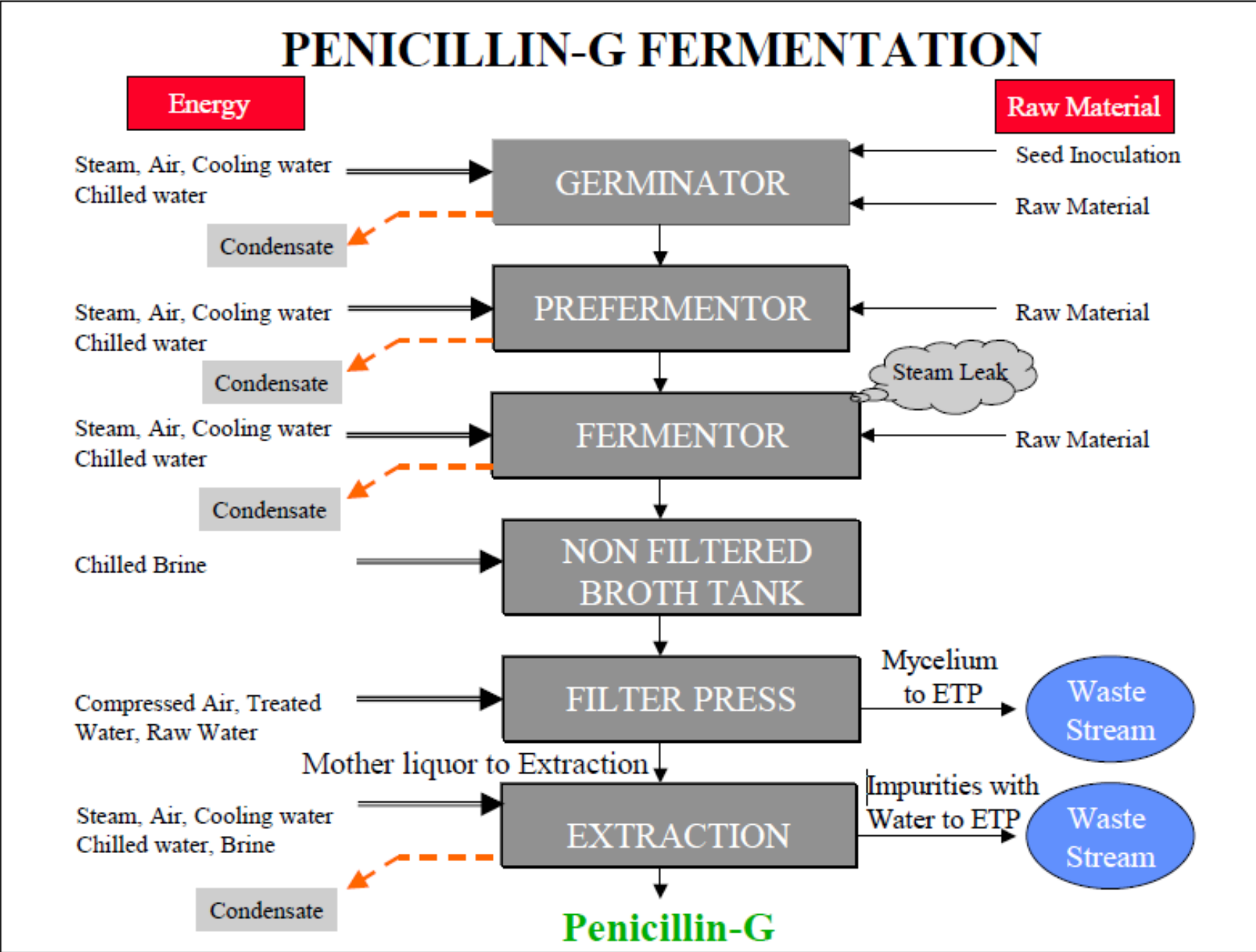
It is important to plan additional data gathering carefully. Here are some basic tips to avoid wasting time and effort:

- **measurement systems should be easy to use and provide the information to the accuracy** that is needed, not the accuracy that is technically possible
- measurement equipment can be **inexpensive** (flow rates using a bucket and stopwatch)
- the quality of the data must be such that the **correct conclusions are drawn** (what grade of product is on, is the production normal etc)
- define how frequent data collection should be to account for process variations.
- measurement exercises over abnormal workload periods (such as startup and shutdowns)
- design values can be taken where measurements are difficult (cooling water through heat exchanger)

Draw process flow diagram and list process steps; identify waste streams and obvious energy wastage

An overview of unit operations, important process steps, areas of material and energy use and sources of waste generation should be gathered and should be represented in a flowchart as shown in the figure below. Existing drawings, records and shop floor walk through will help in making this flow chart. Simultaneously the team should identify the various inputs & output streams at each process step.

Example: A flowchart of Penicillin-G manufacturing is given in the figure below.



Note that waste stream (Mycelium) and obvious energy wastes such as condensate drained and steam leakages have been identified in this flow chart

The audit focus area depends on several issues like consumption of input resources, energy efficiency potential, impact of process step on entire process or intensity of waste generation / energy consumption. In the above process, the unit operations such as germinator, pre-fermentor, fermentor, and extraction are the major conservation potential areas identified.

Identification of Energy Conservation Opportunities

- **Fuel substitution:** Identifying the **appropriate fuel** for efficient energy conversion
- **Energy generation :**Identifying **Efficiency opportunities in energy conversion equipment/utility** such as captive power generation, steam generation in boilers, thermic fluid heating, optimal loading of DG sets, minimum excess air combustion with boilers/thermic fluid heating, optimising existing efficiencies, efficient energy conversion equipment, biomass gasifiers, Cogeneration, high efficiency DG sets, etc.
- **Energy distribution:** Identifying Efficiency opportunities network such as transformers, cables, switchgears and power factor improvement in electrical systems and chilled water, cooling water, hot water, compressed air, Etc.
- **Energy usage by processes:** This is where the major opportunity for improvement and many of them are hidden. Process analysis is useful tool for process integration measures.

Technical and Economic feasibility

The technical feasibility should address the following issues

- Technology availability, space, skilled manpower, reliability, service etc
- The impact of energy efficiency measure on safety, quality, production or process.
- The maintenance requirements and spares availability

The Economic viability often becomes the key parameter for the management acceptance. The economic analysis can be conducted by using a variety of methods. Example: Pay back method, Internal Rate of Return method, Net Present Value method etc. For low investment short duration measures, which have attractive economic viability, simplest of the methods, payback is usually sufficient.

Sample Worksheet for Economic Feasibility

Name of Energy Efficiency Measure

1. Investment

- Equipments
- Civil works
- Instrumentation
- Auxiliaries

2. Annual operating costs

- Cost of capital
- Maintenance
- Manpower
- Energy
- Depreciation

3. Annual savings

- Thermal Energy
- Electrical Energy
- Raw material
- Waste disposal

Net Savings /Year (Rs./year)
= (Annual savings-annual operating costs)

Payback period in months
= (Investment/net savings/year) x 12

Energy Audit Reporting Format

After successfully carried out energy audit energy manager/energy auditor should report to the top management for effective communication and implementation.

A typical energy audit reporting contents and format are given below. The following format is applicable for most of the industries. However the format can be suitably modified for specific requirement applicable for a particular type of industry.

Report on

DETAILED ENERGY AUDIT

TABLE OF CONTENTS

i. Acknowledgement

ii. Executive Summary

Energy Audit Options at a glance & Recommendations

1.0 Introduction about the plant

1.1 General Plant details and descriptions

1.2 Energy Audit Team

1.3 Component of production cost (Raw materials, energy, chemicals, manpower, overhead, others)

1.4 Major Energy use and Areas

2.0 Production Process Description

2.1 Brief description of manufacturing process

2.2 Process flow diagram and Major Unit operations

2.3 Major Raw material Inputs, Quantity and Costs

3.0 Energy and Utility System Description

3.1 List of Utilities

3.2 Brief Description of each utility

3.2.1 Electricity

3.2.2 Steam

3.2.3 Water

3.2.4 Compressed air

3.2.5 Chilled water

3.2.6 Cooling water

4.0 Detailed Process flow diagram and Energy& Material balance

4.1 Flow chart showing flow rate, temperature, pressures of all input-output streams

4.2 Water balance for entire industry

5.0 Energy efficiency in utility and process systems

5.1 Specific Energy consumption

5.2 Boiler efficiency assessment

5.3 Thermic Fluid Heater performance assessment

5.4 Furnace efficiency Analysis

5.5 Cooling water system performance assessment

5.6 DG set performance assessment

5.7 Refrigeration system performance

5.8 Compressed air system performance

5.9 Electric motor load analysis

5.10 Lighting system

6.0 Energy Conservation Options & Recommendations

6.1 List of options in terms of No cost/ Low Cost, Medium cost and high investment Cost, Annual Energy & Cost savings, and payback

6.2 Implementation plan for energy saving measures/Projects

ANNEXURE

A1. List of Energy Audit Worksheets

A2. List of instruments

A3. List of Vendors and Other Technical details

Benchmarking and Energy Performance

Benchmarking of energy consumption internally (historical / trend analysis) and externally (across similar industries) **are two powerful tools for performance assessment and logical evolution of avenues for improvement.** Historical data well documented helps to bring out energy consumption and cost trends month-wise / day-wise. Trend analysis of energy consumption, cost, relevant production features, specific energy consumption, help to understand effects of capacity utilization on energy use efficiency and costs on a broader scale.

External benchmarking relates to inter-unit comparison across a group of similar units. However, it would be important to ascertain similarities, as otherwise findings can be grossly misleading. Few comparative factors, which need to be looked into while benchmarking externally are:

- Scale of operation
- Vintage of technology
- Raw material specifications and quality
- Product specifications and quality

Benchmarking energy performance permits

- Quantification of fixed and variable energy consumption trends vis-à-vis production levels
- Comparison of the industry energy performance with respect to various production levels (capacity utilization)
- Identification of best practices (based on the external benchmarking data)
- Scope and margin available for energy consumption and cost reduction
- Basis for monitoring and target setting exercises.

The benchmark parameters can be:

- Gross production related

e.g. kWh/MT clinker or cement produced (cement plant)

e.g. kWh/kg yarn produced (Textile unit)

e.g. kWh/MT, kCal/kg, paper produced (Paper plant)

e.g. kCal/kWh Power produced (Heat rate of a power plant)

e.g. Million kilocal/MT Urea or Ammonia (Fertilizer plant)

e.g. kWh/MT of liquid metal output (in a foundry)

- Equipment / utility related

e.g. kW/ton of refrigeration (on Air conditioning plant)

e.g. % thermal efficiency of a boiler plant

e.g. % cooling tower effectiveness in a cooling tower

e.g. kWh/NM³ of compressed air generated

e.g. kWh /litre in a diesel power generation plant.

While such benchmarks are referred to, related crucial process parameters need mentioning for meaningful comparison among peers. For instance, in the above case:

- **For a cement plant** – type of cement, blaine number (fineness) i.e. Portland and process used (wet/dry) are to be reported alongside kWh/MT figure.
- **For a textile unit** – average count, type of yarn i.e. polyester/cotton, is to be reported along side kWh/square meter.
- **For a paper plant** – paper type, raw material (recycling extent), GSM quality is some important factors to be reported along with kWh/MT, kCal/Kg figures.
- **For a power plant / cogeneration plant** – plant % loading, condenser vacuum, inlet cooling water temperature, would be important factors to be mentioned alongside heat rate (kCal/kWh).
- **For a fertilizer plant** – capacity utilization(%) and on-stream factor are two inputs worth comparing while mentioning specific energy consumption

- **For a foundry unit** – melt output, furnace type, composition (mild steel, high carbon steel/cast iron etc.) raw material mix, number or power trips could be some useful operating parameters to be reported while mentioning specific energy consumption data.
- **For an Air conditioning (A/c) plant** – Chilled water temperature level and refrigeration load (TR) are crucial for comparing kW/TR.
- **For a boiler plant** – fuel quality, type, steam pressure, temperature, flow, are useful comparators alongside thermal efficiency and more importantly, whether thermal efficiency is on gross calorific value basis or net calorific value basis or whether the computation is by direct method or indirect heat loss method, may mean a lot in benchmarking exercise for meaningful comparison.
- **Cooling tower effectiveness** – ambient air wet/dry bulb temperature, relative humidity, air and circulating water flows are required to be reported to make meaningful sense.
- **Compressed air specific power consumption** – is to be compared at similar inlet air temperature and pressure of generation.
- **Diesel power plant performance** – is to be compared at similar loading %, steady run condition etc.